Corrosion Science 191 (2021) 109711

Contents lists available at ScienceDirect

Corrosion Science

journal homepage: www.elsevier.com/locate/corsci

Effect of Al content on the thermal oxidation behaviour of AlHfMoNbTi high-entropy alloys analysed by *in situ* environmental TEM

Xiaomeng Yang, Zibing An, Yadi Zhai *, Xin Wang, Yanhui Chen *, Shengcheng Mao, Xiaodong Han

Institute of Microstructure and Property of Advanced Materials, Beijing University of Technology, Beijing, 100124, China

ARTICLE INFO

Keywords:
Oxidation
Refractory high-entropy alloy
Al addition
In situ
Environmental TEM

ABSTRACT

The high-temperature oxidation behaviours of AlxHfMoNbTi-series refractory high-entropy alloys with an Al content ranging from 0 to 20 at.% (x=0-1) were tested at 700–1000 °C. An *in situ* oxidation experiment was performed at 25–900 °C using environmental transmission electron microscopy (ETEM) to reveal the oxidation mechanism. The addition of Al to the HfMoNbTi alloy effectively increased the alloy strength and hardness. The improvement in oxidation resistance by Al addition is attributed to the formation of mixed oxide structures instead of layered oxides. Mo-rich precipitates are preferentially oxidized, and their volatilization at the metal/oxide interface accelerates alloy oxidation.

1. Introduction

High-entropy alloys (HEAs) have recently received extensive attention due to their excellent high-temperature mechanical properties, resistance to radiation damage, wear resistance, and corrosion resistance. Refractory high-entropy alloys (RHEAs) are widely developed due to their excellent high-temperature mechanical properties and are a potential alternative alloy to superalloys. RHEAs can bear higher service temperatures than conventional superalloys, which usually soften above 1000 °C. It has been reported that HfMoNbTaTiZr can endure a compressive strength of 556 MPa at 1200 °C [1] and that NbMoTaVW can withstand 477 MPa at 1600 °C [2]. A high content of refractory elements has been shown to greatly improve the high-temperature mechanical properties of RHEAs. Many series of high-performance RHEAs have been designed; for example, HfMoNbTiZr, HfMoTaTiZr, and $CrMo_{0.5}NbTa_{0.5}TiZr$ all exhibit good mechanical properties, especially at high temperatures, and all of their high-temperature strengths are higher than those of Inconel 718 at 1000 °C [3–7].

A serious issue, however, lies in the fact that all of these bare alloys undergo severe oxidation at high temperatures beyond 800 °C in service [8,9]. Oxidation corrodes surface metals, diminishes the mechanical properties and decreases the service life, even causing overall alloy destruction if oxidation-resistant elements or protective layers are lacking [10]. Some studies have been performed to improve the oxidation resistance of RHEAs [11–15], and adding specific alloying elements is an effective method to improve the corrosion performance of alloys. Additions of Al, Cr and Si can usually form protective oxide films such as Al_2O_3, Cr_2O_3 and SiO_2, respectively, to prevent further oxidation of base metals in corrosion protective science [16]. During the development of conventional antioxidation alloys, fast improvement in corrosion resistance is usually based on a deep understanding of the oxidation mechanism [17]. Forming protective oxide scales on the surface is an effective method for conventional stainless steel, and rapid diffusion of enough Cr atoms into the surface is essential during this process. HEAs usually exhibit a very slow diffusion speed, and the diffusion of Al is estimated to be different [18]. A deep understanding of the role of specific elements on the oxidation behaviour and mechanism is important for the development of advanced HEAs that operate in harsh environments [19].

Aluminium is the most effective antioxidant additive in conventional alloys since it can be oxidized to form a dense Al_2O_3 oxide layer that protects the base alloy. Furthermore, the addition of Al can decrease the density of the alloy and enhance its strength [20]. The addition of Al into HEAs has been reported in HfNbTiZr, which indicated that Al can protect the alloy and lead to the least precipitation in the alloy under oxidation conditions [19]. The mechanism of Al addition to HEAs has been reported in 3d transition-metal HEAs, and Al can be selectively oxidized as in stainless steel or NiAl alloys. Increasing the Al content protects the alloy by promoting the formation of a more complete protective film [19,20]. Al- and Nb-containing RHEAs can form $AlNbO_4$ composite oxides at high temperatures to protect the inner alloy [19,

* Corresponding authors.
E-mail addresses: ydzhai@bjut.edu.cn (Y. Zhai), yhchen@bjut.edu.cn (Y. Chen).

https://doi.org/10.1016/j.corsci.2021.109711
Received 14 March 2021; Received in revised form 18 July 2021; Accepted 20 July 2021
Available online 28 July 2021
0010-938X/© 2021 Elsevier Ltd. All rights reserved.

21]. Microstructure and elemental distribution variation during oxidation provide a full picture of the oxidation process and offer direct clues regarding oxidation mechanisms. However, studies on the microstructure of the oxide layer [22,23] in RHEAs are limited. Furthermore, there is no intuitive evidence or analysis of the oxidation mechanism, especially for RHEAs, so there is only speculation from posterior experiments, which cannot clarify the specific roles of the elements. Conventional scanning electron microscopy (SEM) and transmission electron microscopy (TEM) are used to analyse the structure of oxide layers after oxidation [16,23]. The recent development of environmental transmission electron microscopy (ETEM) has given us the opportunity to study solid/gas interactions *in situ* and has been applied to many materials; *in situ* data during oxidation will provide direct evidence of the oxidation mechanism [24,25].

In this study, the oxidation behaviours of $Al_xHfMoNbTi$ entropy alloys with an Al content ranging from 0 to 20 at% were studied at 700 °C–1000 °C. The evolution of the morphology and microstructure was analysed as a function of Al content. The effect of Al addition and the oxidation mechanism were investigated by analysing the cross-section of the oxide layer and performing *in situ* ETEM experiments. This research on oxidation behaviours provides direct data for the development and design of new antioxidation HEAs.

2. Experimental details

$HfMoNbTiAl_x$ (x = 0, 0.25, 0.5, 0.75, and 1), hereafter denoted as Al_x, was prepared by arc melting in a water-cooled copper crucible. The pure metal particles were of 99.9 % purity. Each ingot was flipped over and remelted at least eight times to ensure homogeneity of the elements, and all samples were tested under as-cast conditions. Compression samples 3 mm in diameter and 4.5 mm in height were prepared by wire cutting to test their mechanical properties. Samples used for the oxidation test were cut into squares with an edge length of 4 mm. The samples were then subjected to thermal oxidation in a muffle furnace under ambient conditions at temperatures from 700 °C to 1000 °C. The Vickers hardness of the alloys was measured using an indentation load of 200 g, and the mean value of three measurements was taken as the hardness.

A BRUKER-D8 ADVANCE X-ray diffraction (XRD) system with a Cu Kα radiation source was used to measure the crystallographic structures of the samples, and data were collected in a 2θ range of 20–100° at a scanning rate of 0.02° s^{-1}. Mechanically polished cross-sections of the oxidized samples were then characterized by an FEI Quanta 650 scanning electron microscopy (SEM) system equipped with energy dispersive X-ray spectroscopy (EDS). The oxidized surface layers on the oxidized samples were extracted using an FEI Helios 600i focused ion beam (FIB) and then studied using an FEI Themis transmission electron microscopy (TEM) system at 300 kV. EDS and elemental mapping were performed by electron dispersive X-ray spectroscopy (Oxford) equipped within the TEM system. K series peaks for O, Al and Ti and L series peaks for Hf, Mo and Ta were used to quantify the measurements. Thin-film samples from dendritic arms were prepared by the FIB technique for *in situ* oxidation tests. The samples were then transferred to a microelectromechanical system (MEMS, BestronST) heating chip and heated from 20 °C to 900 °C in an environmental transmission electron microscopy (FEI Titan ETEM) system operated at 300 kV with a pure oxygen pressure of 0.5 mbar.

3. Results and discussions

Fig. 1 shows the XRD pattern of the as-cast alloys and typical TEM images and electron diffraction patterns (EDPs). As known from the ICSD card, both the body-centred cubic (BCC) phase and B2 phase have the same lattice parameters, with the overlapping of some peaks such as 110 and 200. The B2 phase, which has ordered structures, has its own distinguishing peaks as 100 and 111 peaks. The Al_0 alloy exhibits a single-phase BCC structure, while $Al_{0.25}$ alloys also contain the B2 phase in some grains identified by TEM and diffraction patterns, as shown in the right part of Fig. 1. Al addition prompted precipitation of the B2 structure, as reported [26]. In alloys with a high Al ($Al_{0.50}$ and $Al_{0.75}$), all the phases convert to the B2 phase. Typical diffraction patterns of Al_0, $Al_{0.25}$ (BCC + B2), and $Al_{0.5}$ (B2 only) are shown in Fig. 1b. The BCC lattice constants of the alloys are calculated based on Bragg's formula, as presented in Fig. 1b. With increasing Al content, the lattice constant gradually decreases from 0.3301 nm to 0.3258 nm. The decrease in the lattice constant can be explained by the increasing Al content, as Al has the smallest atomic diameter of 118 pm among the constituents, *e.g.*, Hf (208 pm), Mo (190 pm), Nb (198 pm) and Ti (167 pm).

All five original alloys have similar microstructures, and the microstructure and elemental distribution of a typical $Al_{0.5}$ alloy are shown in Fig. 2. This figure shows typical backscattered electron images of the as-cast $Al_{0.5}$ alloys and EDS mapping. All five original alloys exhibit coarse dendrite structures. Elements with relatively low melting temperatures, such as Al (660 °C), Ti (1668 °C) and Hf (2233 °C), are prone to segregate at the interdendritic region, and the dendrite arm is rich in Mo

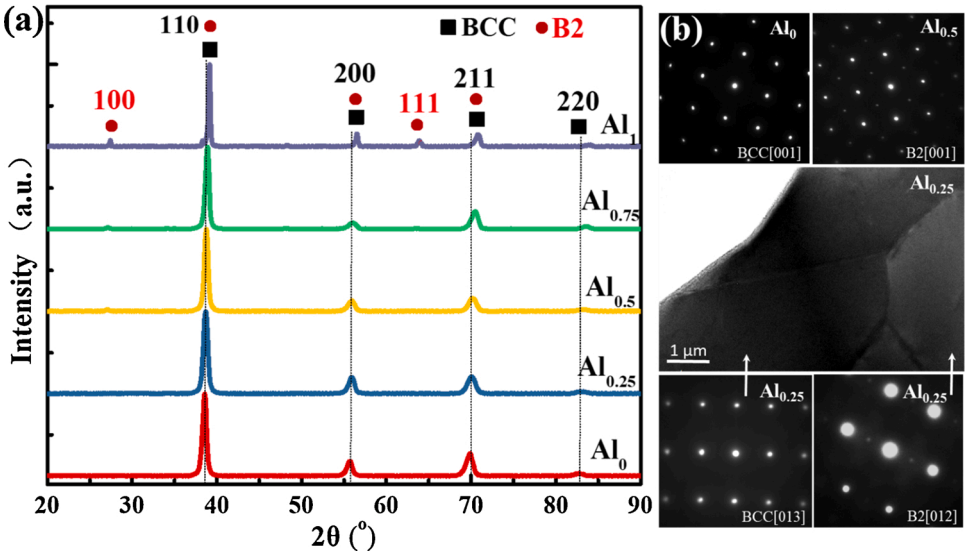


Fig. 1. (a) X-ray diffraction patterns and lattice parameters of the original alloys and their lattice constants and (b) typical TEM images and electron diffraction patterns.

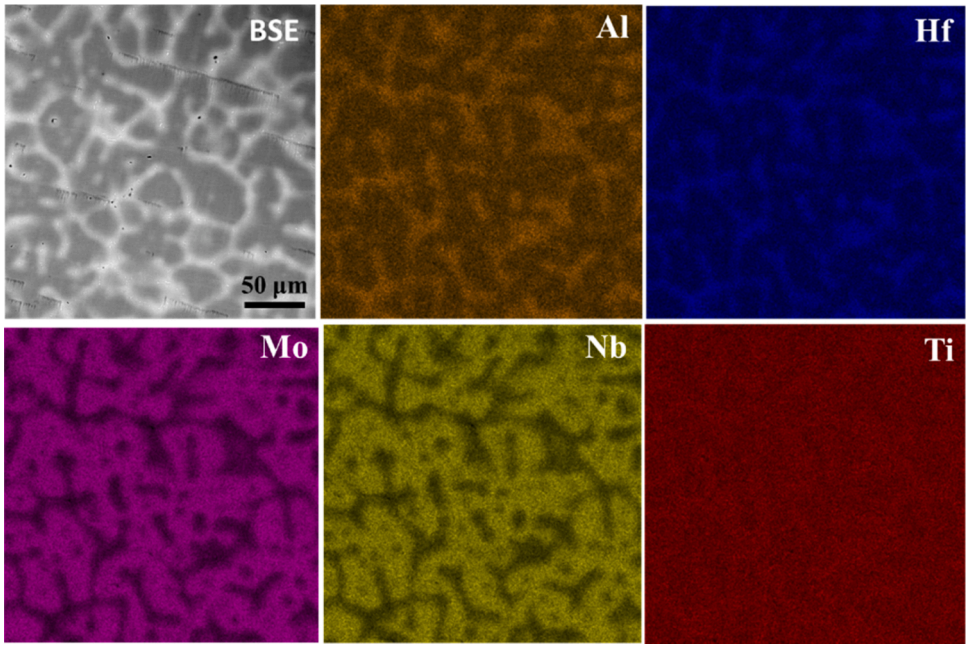


Fig. 2. SEM backscattering electron images of the $Al_{0.5}$HfMoNbTi alloy and EDS mapping.

(2620 °C) and Nb (2470 °C).

The compressive strength and hardness at room temperature are shown in Fig. 3a. The initial Al_0 alloy has a strength of 1280 MPa, and the maximum plastic strain is 6%. The strengths of $Al_{0.25}$ and $Al_{0.5}$ increased to 1500 MPa after a small amount of Al was added. At the same time, the brittleness increased with increasing Al content. As the Al content increased, the room temperature brittleness of the alloy continued to increase because the addition of Al caused the whole alloy to transform into a B2 structure. Further increasing the Al content makes the 20 at.% Al_1 alloy too brittle to be cut by electrical discharge wire cutting for the compressive test. The average Vickers hardness of the five alloys in Fig. 3b indicates that the Vickers hardness is also greatly improved with the addition of Al. A small amount of Al can yield the maximum improvement rate, and a continuous increase in the Al content only slowly increases the hardness of the alloy after the Al content is higher than 5 at.%.

The oxidation kinetics curves of all alloys were tested at 1000 °C in laboratory air. The mass gain-time curve is shown in Fig. 4a. The oxidation kinetics curve can be roughly divided into two stages: the early stage of mass increase and the long-term stage of mass loss. During the oxidation process, within 5 h, the mass gain does not show much difference with different Al contents. After 5 h, all alloy samples show varying degrees of mass loss. It is known that MoO_3 will volatilize at temperatures higher than 700 °C; our alloy contains 20 at.% to 25 at.% Mo, and the mass loss can be attributed to the evaporation of MoO_3 [27]. However, if the protective, dense Al_2O_3 layer is thick enough, evaporation can be prohibited, and mass loss will be retarded. This retardation effect can be seen in Al-containing alloys after oxidation for 5 h, and all the mass gain curves show a downward trend. When the Al content is higher than 0.75, the mass remains constant, which indicates that the formed oxide layer has a good retardation effect. However, the Al_1 alloy cracked in the subsequent process due to its enhanced brittleness causing mass loss. All five alloys have similar oxidation products, and the typical XRD pattern of $Al_{0.5}$ is shown in Fig. 4b, mainly containing HfO_2, Nb_2O_5, TiO_2, $TiNb_2O_7$ and $HfTiO_4$.

The cross-sectional images of different alloys oxidized at 1000 °C for 1 h are shown in Fig. 5. The alloy without Al contains a large number of cracks in the Mo-rich layer, indicating a greater tendency to spall than the alloy containing Al. The addition of Al significantly reduced the thickness of the Mo-rich layer and the number of cracks. However, the thickness of the oxide layer is not independently controlled by the content of Al and Mo. The $Al_{0.5}$ alloy has the thinnest oxide layer and the lowest oxidative weight gain at 1 h. The oxide layer of the $Al_{0.75}$ alloy is relatively dense, and although cracks are visible inside it, there are almost no holes, which corresponds to its increasing oxidation mass gain, indicating that the oxide layer is more protective than other alloys.

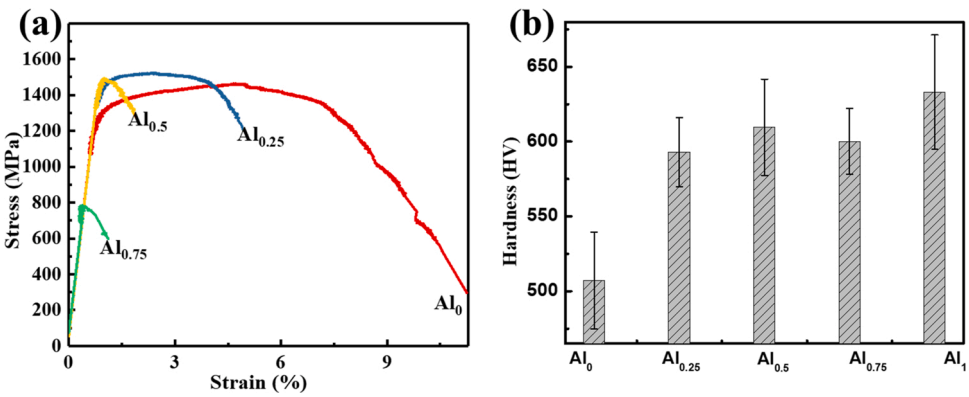


Fig. 3. Mechanical properties of AlHfMoNbTi alloys with different Al contents: (a) compressive strain-stress relation and (b) Vickers hardness.

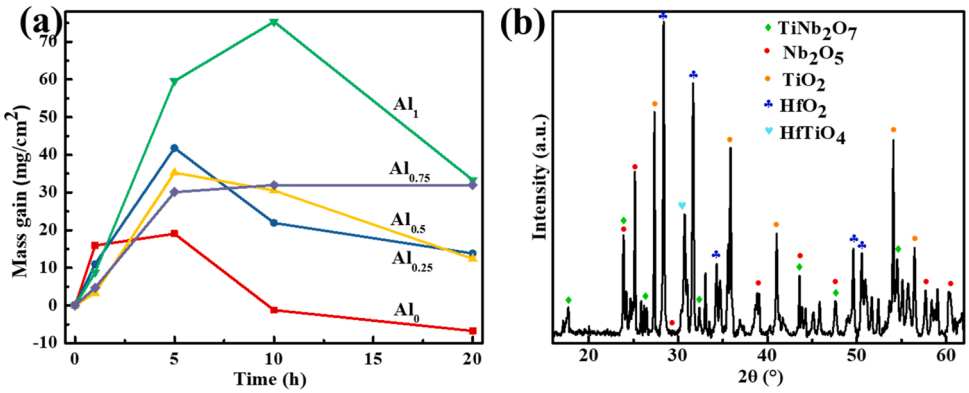


Fig. 4. Oxidation after heating for 20 h at 1000 °C: (a) mass gain of the five alloys and (b) XRD pattern of the oxidized $Al_{0.5}$ alloy after 20 h.

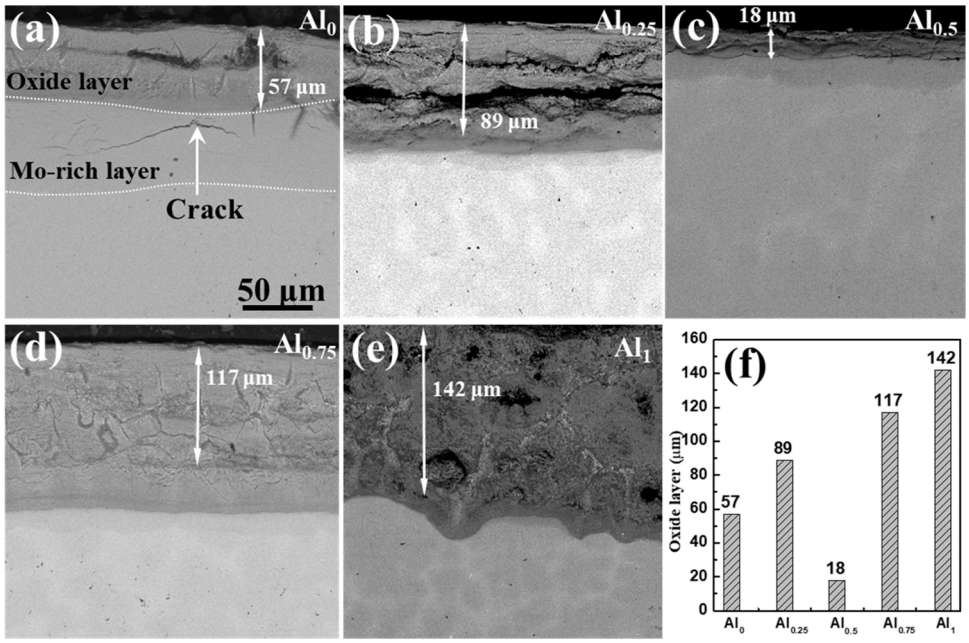


Fig. 5. Cross-section of alloys oxidized at 1000 °C for 1 h, (a) Al_0, (b) $Al_{0.25}$, (c) $Al_{0.5}$, (d) $Al_{0.75}$, and (e) Al_1, and their thickness difference.

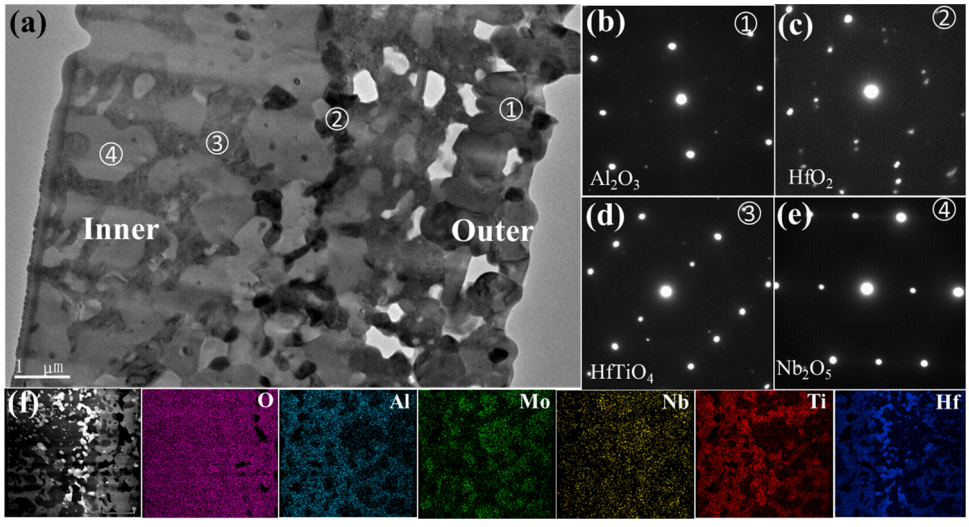


Fig. 6. Surface of the $Al_{0.25}$ alloy after oxidation for 30 min at 1000 °C. (a) TEM image, (b-e) SAED patterns in regions 1-4, and (f) HAADF image and EDS mapping.

TEM lamellae from the oxide layer of alloys oxidized at 1000 °C for 0.5 h were cut and removed by the FIB technique to better understand the products after the initial stage of oxidation. The oxides of the $Al_{0.25}$, $Al_{0.5}$ and $Al_{0.75}$ alloys have similar morphologies, and typical TEM images and selected area electron diffraction (SAED) patterns of $Al_{0.25}$ are shown in Fig. 6. Four typical regions (indexed as 1–4 in Fig. 6a) were selected, and their structures were identified using SAED. The outer layer forms faster than the inner layer, and the SAED pattern shows that region 1 contains Al_2O_3 (−423.5 J/mol) and that region 2 contains HfO_2 (−442.26 J/mol), which both have lower oxide formation energy than other oxides in the system and form first at the top several hundred nanometres of the alloy. Region 3 is a complex $HfTiO_4$ oxide and exhibits a tortuous shape due to the reaction of equiaxed elemental oxide with adjacent crystals. Since the sample is tested at 1000 °C, MoO_3 will be seriously volatilized above 700 °C, and it is difficult to determine whether its formation and volatilization have an effect on the oxidation behaviour. *In situ* experiments were carried out to obtain all structural and elemental variations to illuminate the effect of Al and Mo during oxidation.

An *in situ* oxidation experiment was performed in an ETEM chamber with a gas pressure of 0.5 mbar to better understand the initial oxidation process and how the oxide layer formed. $Al_{0.5}$ and Al_1 alloys were chosen to carry out *in situ* experiments from 20 °C to 900 °C with a temperature interval of 50 °C for 5 min, and only images under key temperatures are shown in Fig. 11. Typical images of the $Al_{0.5}$ oxidation process are given in Fig. 7. Stress streaks induced in the FIB cutting process [28] are visible from room temperature to 300 °C, and further elevated thermal oxidation at 400 °C can completely eliminate such stress [29]. The initial obvious oxidation stage starts at 700 °C–750 °C, and some tiny black oxide particles begin to grow at the edge of the sample and continue to oxidize forward into the inner alloy with increasing time. Unlike the performance of traditional simple substances or alloys [30,31], oxides hardly undergo epitaxial growth in this alloy, and the sample boundaries hardly change during the oxidation process. This fact indicates that the internal diffusion of the oxidation mode dominates the entire oxidation process. When the temperature rises above 850 °C, the alloy lamella begins to bend due to the stress caused by thermal expansion.

A typical high-angle annular dark-field (HAADF) image after *in situ* oxidation at 900 °C is presented in Fig. 8a, and there are obvious holes in the sample border. Electron energy loss spectroscopy (EELS) was performed to determine the nanoscale structure and elemental composition of oxides on the edges. Since the contrast of HAADF imaging mainly depends on the atomic number Z, different contrasts are used to distinguish between oxides. Three types of particles with typical contrast, namely, particles with bright white contrast, grey particles, and black particles, are selected from left to right, marked with red circles and indexed with numbers. The region in circle 1 mainly contains Nb and Ti oxides with the formula $Nb_{39.9}Ti_{7.4}Hf_{2.6}O_{50.1}$, circle 2 mainly contains Al and Ti oxides with the formula $Al_{22.5}Ti_{23.3}Hf_{17.2}O_{37}$, and circle 3 mainly contains oxides with the formula $Hf_{35}Ti_3Pt_{23}O_{39}$. The Pt edge appeared because the FIB technique used Pt sputtering. The results indicate that Hf oxide formed first, Ti and Al oxides formed next, and Nb oxide formed last with a slow oxide growth rate. None of the spectra detected the presence of Mo in the outermost layer, and Mo was not observed in the oxide layer, as shown in the SEM images in Fig. 5. Therefore, the formation order of oxides other than Mo oxides was

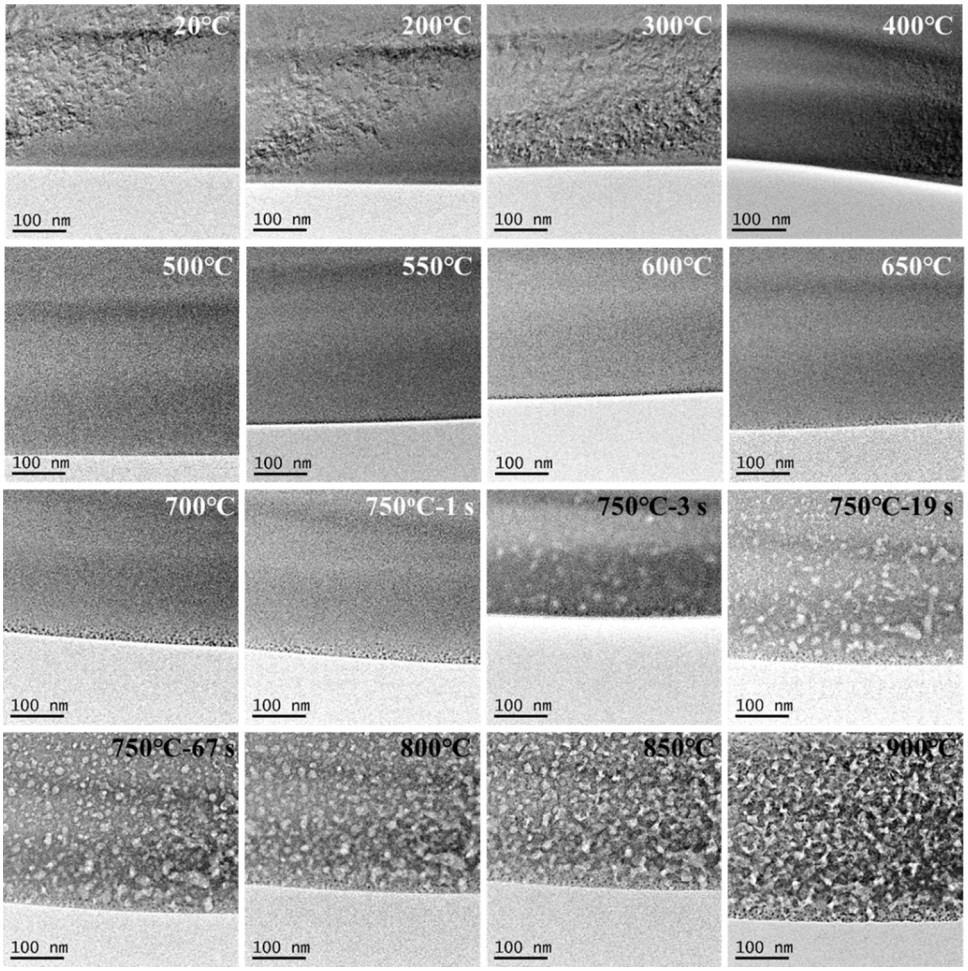


Fig. 7. The *in situ* oxidation process of the $Al_{0.5}$ alloy under 0.5 mbar oxygen from 20 to 900 °C is shown in supplementary video file V1.

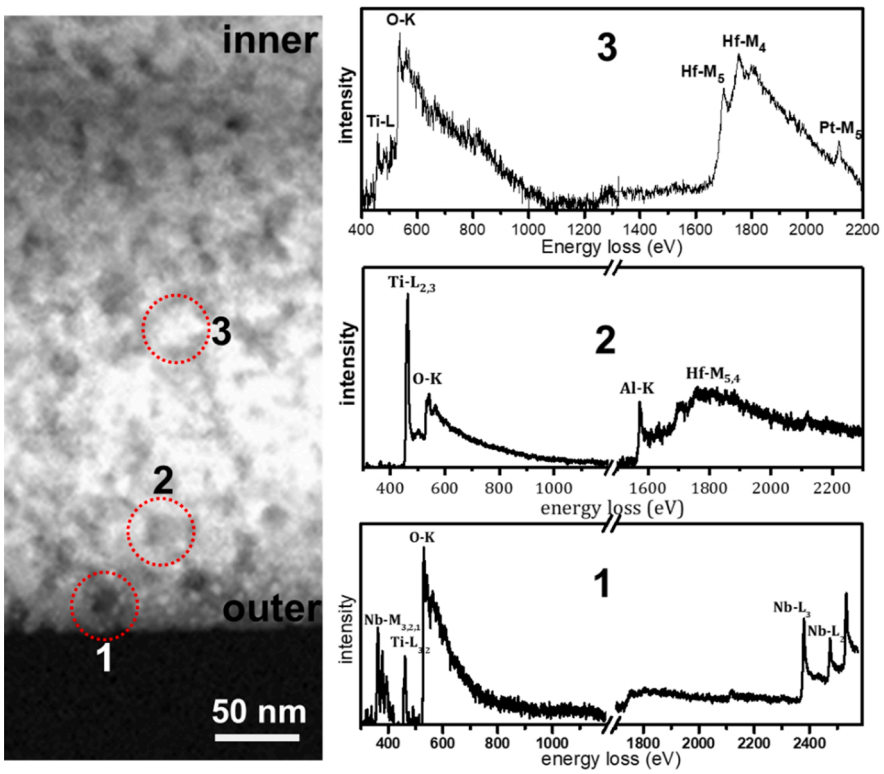


Fig. 8. HAADF image and typical EELS spectra after *in situ* oxidation, with oxides marked in red circles and indexed with numbers.

determined according to *in situ* experiments.

It is known in bulk and our other *in situ* experiments that $Al_{0.5}$ has a high initial oxidation temperature, and Al_1, with the lowest initial oxidation temperature, is used to reduce the volatilization of MoO_3. Al_1 was then used to determine the role of Mo in the oxidation process. An *in situ* oxidation process of Al_1 is shown in Fig. 9 from room temperature to 900 °C to obtain all oxide products. There was no obvious structural variation at temperatures below 650 °C. Most alloys were amorphized, and only some particles existed after heating for 15 min at 450 °C. Some black particles formed when the temperature increased to 650 °C and continued to grow into band-like structures hundreds of nanometres wide. The experiment was stopped after heating at 650 °C to ensure that a large amount of MoO_3 would not be volatilized. EDS point analysis shows that the black rod-shaped particles in area 1 (Fig. 9) are rich in Mo; they have a composition of $Mo_{20}M_{24}O_{56}$ in region 1, where M indicates other metals. Areas with white contrast appear adjacent to particles with black contrast and continue to expand (10 s), which indicates that stress appears during the growth process. Grey regions appear between black and white particles and continue to grow, with a composition of $Mo_{13}M_{30}O_{57}$, as indexed in region 2.

The oxidation sequence is determined to be Mo>Hf > Ti and Al > Nb, indicating that Mo has the highest diffusion rate and preferentially precipitates to form oxides, which also corresponds to the appearance of Mo-enriched layers in bulk samples. The formation of Mo oxides leads to the depletion of nearby Mo and enrichment of other elements. Other oxides will nucleate when the concentration is sufficient. The second growth process of Mo oxide particles in the experiment may be due to the surrounding elements also turning into oxides, which locally lose the sluggish diffusion effect, so Mo in other regions can diffuse to the existing particles and continue to grow; this process does not happen at higher temperatures (>750 °C)

SAED patterns were collected at different temperatures during the *in situ* experiment of the Al_1 alloy to determine the structure of the oxidation products, as shown in Fig. 10. When the sample was heated to 450 °C, the sample was tilted due to uneven thermal expansion of Pt welded at the back of the sample, resulting in diffraction spots appearing in the vertical direction. However, during the 450 °C insulation process, the alloy underwent an amorphization stage during heating, diffraction spots gradually disappeared, and amorphous rings appeared. After 15 min, almost all the diffraction spots disappeared, and amorphous halos began to appear. When the temperature was dropped from 450 °C to 200 °C, after holding for 10 min, some diffraction spots reappeared, and the distance between the (200) planes was found to be the same as that of the original alloy; that is, the reduced temperature can restore the amorphized sample to its original crystalline state. When they were reheated to 500 °C, all samples were found to be amorphous. This result means that the alloy first undergoes a transition from crystalline to amorphous, and then, oxide particles will grow from the amorphous matrix when the oxidation temperature is reached. This amorphization may be caused by the internal diffusion of a large amount of oxygen, resulting in the deformation of the internal crystal lattice, which leads to severe internal oxidation, resulting in large internal stress and causing the oxide layer to peel off. The phenomenon of amorphization still needs further experimental proof.

4. Discussion

4.1. Oxidation sequence

Traditional antioxidation alloys often use selective oxidation to form specific element oxide layers of Al, Cr, Si, *etc.*, to reduce the continuous oxidation of internal alloys. However, in RHEAs, this specific oxide layer is rare, and they all have a large oxidation weight gain. The currently reported RHEA with the best oxidation resistance is AlCrMoNbTi [12]. The mass gain of the Cr-containing alloy AlCrMoNbTi after 48 h is 9 mg/cm^2 at 1000 °C, exhibiting a much better oxidation behaviour than our $Al_{0.5}$ alloy sample, with a mass gain of 30 mg/cm^2 after oxidation for 20 h. In their research, more obvious oxidation stratification can be found. Therefore, it is suggested that the formation of the mixed oxide layer in RHEAs is not conducive to oxidation resistance, and in the

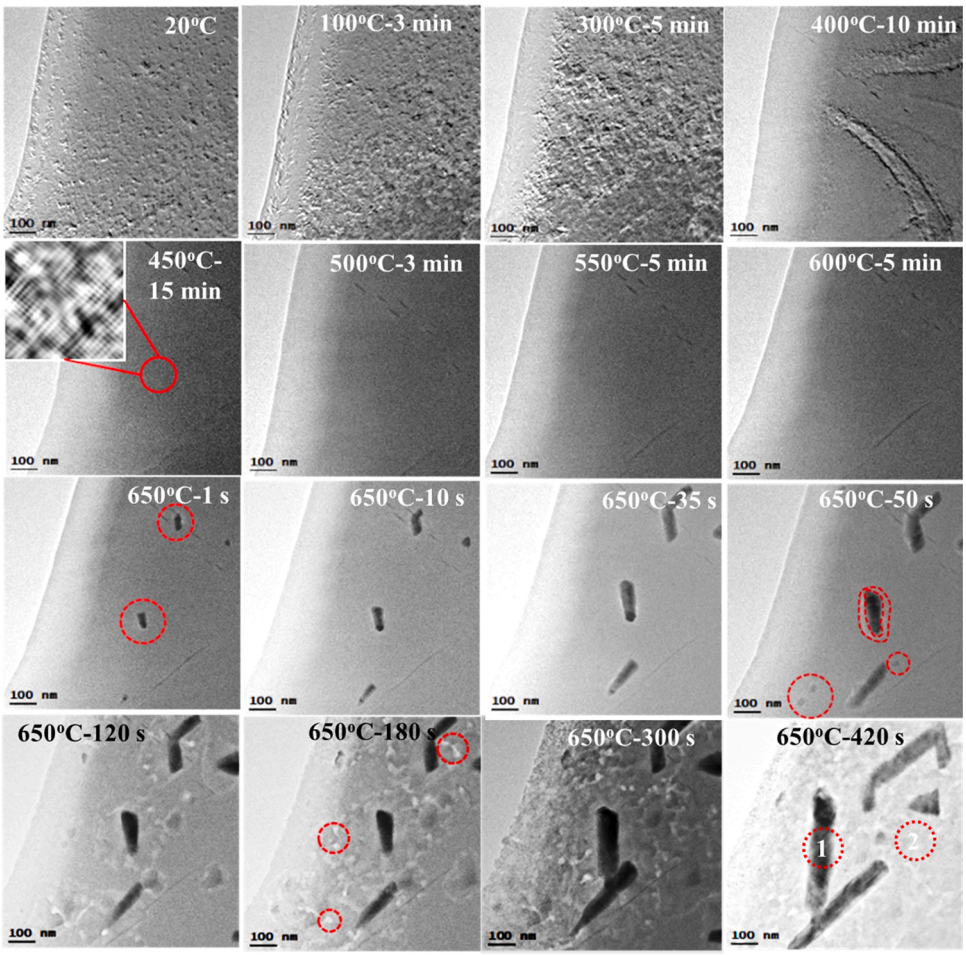


Fig. 9. The *in situ* oxidation process of the Al_1 alloy under 0.5 mbar oxygen from 20 °C to 650 °C is shown in supplementary video file V2.

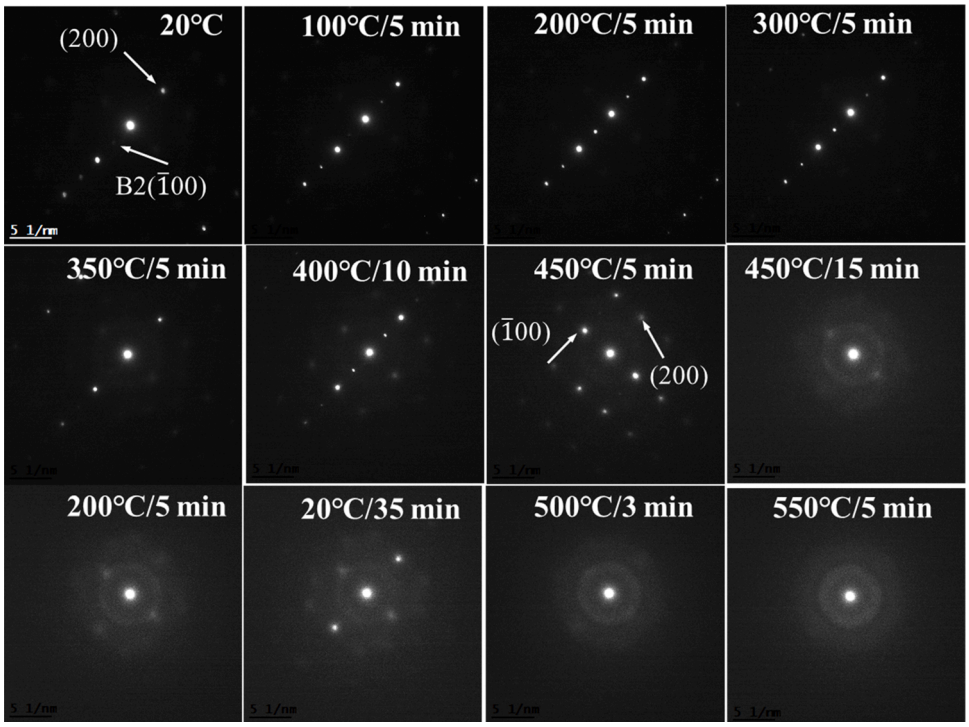


Fig. 10. SAED patterns of Al_1 during *in situ* heating and annealing.

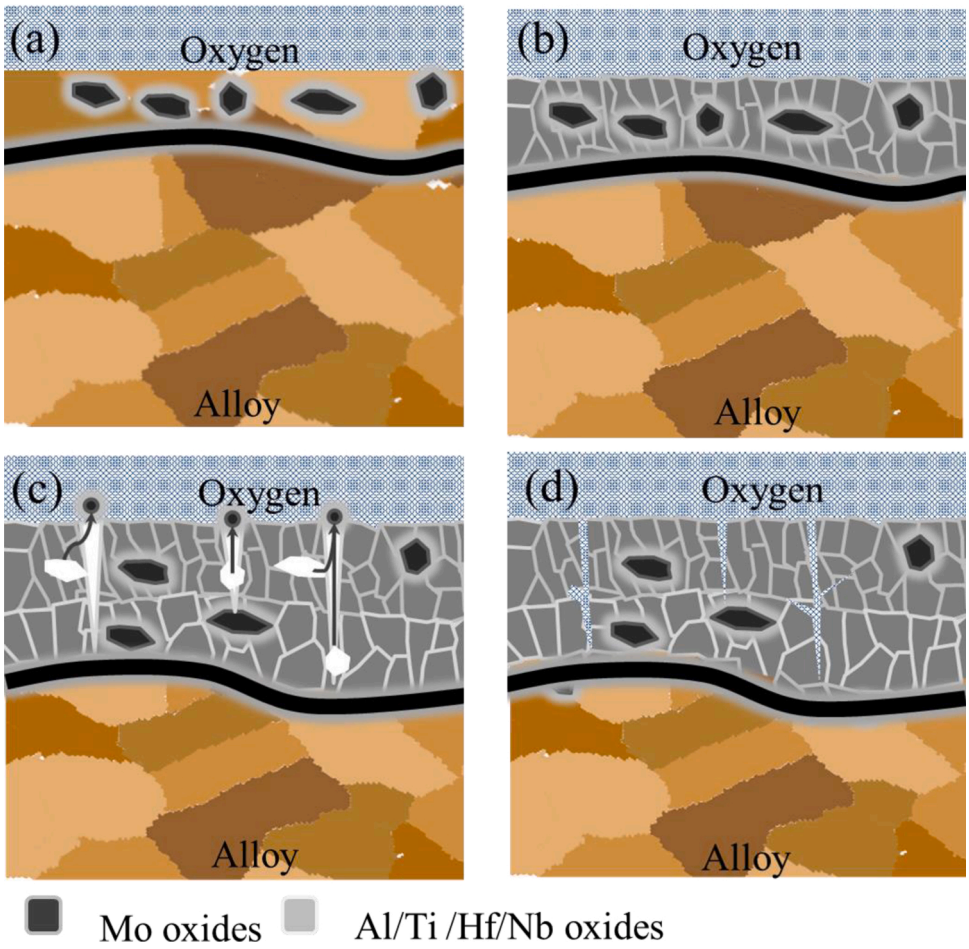


Fig. 11. Schematic of the oxidation mechanism of the AlHfMoNiTi alloy.

present study, this mixed oxide layer cannot hinder the volatilization of Mo oxide, thereby resulting in poor oxidation resistance.

The formation energy of MoO_3 (−146.08 J/mol) is the highest among the five types of metal oxides. The *in situ* experiments show that Mo first forms an oxide when the oxidation temperature is reached and aggregates at the oxide-alloy interface during oxidation of the bulk materials. The violent evaporation of MoO_3 above 800 °C results in the porous structure of the oxide layer, which facilitates the internal diffusion of oxygen. Other elements form oxides at the same time instead of precipitating in sequence to form a specific oxide layer. The formation of various oxides leads to severe volume expansion, which causes the oxide layer to crack. At this time, MoO_3 can volatilize through these cracks to form a porous oxide layer structure. Therefore, in the early stage of oxidation (0−5 h), the oxide layer is thin, and the stress level is not enough to generate cracks. At this time, the oxidation weight gain curve appears to increase. After 5 h, the volatilization of MoO_3 increases due to the generation of cracks in the oxide layer, which makes the weight gain curve show a downward trend.

4.2. Oxidation model

Based on bulk and *in situ* experiments, an oxidation mechanism for this HEA when isothermally oxidized above 700 °C was proposed, as shown in Fig. 11. The oxidation process in our Al-containing alloys is illustrated in Fig. 11. First, as shown in Fig. 11a, oxygen is adsorbed on the alloy surface, and part of the oxygen is dissolved inside the alloy matrix. Mo, which has the fastest diffusion rate in this alloy, is first affected by oxygen to enrich and generate oxides. Then, as shown in Fig. 11b, other elements rarely diffuse, forming a uniformly distributed thin oxide layer. This thin oxide layer is dense but does not hinder the diffusion of oxygen and metal elements, and its growth stress is not enough to cause it to crack and spall. Next, MoO_3 reaches the volatilization temperature and lacks a volatilization pathway. At this time, MoO_3 can still exist in the oxide layer in the solid state. As the thickness of the oxide film increases, the stress level increases, causing a large number of micro-cracks to form in the oxide layer so that MoO_3 can be volatilized, as shown in Fig. 11c. Then, as shown in Fig. 11d, this volatilization causes the oxide layer to become porous, increasing the oxygen invasion path and further deepening oxidation. Therefore, Mo is considered to be the major element leading to poor oxidation resistance.

4.3. Effect of Al addition

Note that the addition of a small amount of Al can improve the obvious oxidation resistance, and the effect is more obvious above 800 °C. Accordingly, the role of Al is analysed. Since the experiment shows that the addition of Al does not form any protective layer in the oxide film, we suggest that the addition of Al results in a decrease in the Mo-rich layer thickness and mitigates different degrees of cracking in the oxide layer. A higher Al content does not mean better oxidation resistance, as a lower Al content (~10 at.%) achieves the best performance. In addition, note that the formation of composite oxides at high temperatures may explain the greater oxidation resistance improvement of high-Al content alloys at high temperatures. It was reported that some composite oxides such as $AlNbO_4$ can be protective [18,32], so some composite oxides were additionally evaluated.

4.4. Expansion after oxidation

The cracking tendency of the oxide layer can be predicted by evaluating the stress levels of the oxide. The volume ratio of the oxide to the metal that forms the oxide, known as the Pilling-Bedworth ratio (PBR), is an important factor in determining the integrity of oxide films [33], and as the main factor that causes growth stress of the oxide film, it can be used to estimate the growth stress of the oxide film. The PBR of the main oxide in this alloy was calculated. Assuming that this alloy oxidizes and diffuses slowly enough at temperatures below 1000 °C, the volume expansion caused by oxidation can be roughly determined by the following equation [34]:

$$\mathrm{PBR} = \frac{V_{oxide}}{V_M} = \frac{W \times n}{\rho \times N \times V_{cell}}$$

In this formula, V_{oxide} is the volume of one mole of the oxide, V_M is the molar volume of the metal in the alloy, W is the molar weight of the oxide, ρ is the density of the oxide, N is Avogadro's number, V_{cell} is the volume of an oxide unit cell, and n is the number of molecules or atoms per unit cell. The result is shown in Table 1. The results show that the formation of Al_2O_3 can significantly reduce the overall PBR of the alloy, that MoO_3 produces considerable volume expansion, and that the PBR slightly increases with the Al content. For composite oxides, the formation of $HfTiO_4$ can reduce the RBR when TiO_2 is consumed in anatase, and it has little effect when TiO_2 is consumed in the rutile structure. The formation of $TiNb_2O_7$ increases the PBR, meaning greater stress generation. An increase in Al content will lead to an overall increase in the RBR of the oxide, but this increase is minor. Assuming that no composite oxide forms, the average PBR of the $Al_{0.5}$ alloy compared to that of the Al_0 alloy decreases by 4.83 %, and that of Al_1 decreases by 9.4 %, which is favourable to prevent the oxide layer from breaking. Note that the stress generated by oxide growth in alloys with a higher Al content is also greater. When all elements are nonuniformly oxidized, alloys with a high Al content will have a greater tendency to spall, which also indicates why a low Al content provides better protection.

Table 1
PBRs for the main oxides in the $Al_xHfMoNbTi$ alloy.

	Al_0	$Al_{0.25}$	$Al_{0.5}$	$Al_{0.75}$	Al_1
Nb_2O_5	1.917	1.935	1.944	1.965	1.994
HfO_2	1.924	1.942	1.951	1.972	1.969
θ-Al_2O_3	1.303	1.315	1.321	1.336	1.356
$HfTiO_4$	1.883	1.900	1.909	1.930	1.958
$TiNb_2O_7$	2.462	2.485	2.496	2.523	2.561
TiO_2(A)	1.895	1.913	1.921	1.942	1.971
TiO_2(R)	1.782	1.798	1.807	1.827	1.854
MoO_3	2.823	2.849	2.862	2.894	2.937

*TiO_2(A) indicates the anatase structure, and TiO_2(R) indicates the rutile structure.

5. Conclusion

The mechanical and oxidation resistance properties of a series of $Al_xHfMoNbTi$ (x = 0–1) HEAs were studied, and their oxidation mechanism was proposed. The conclusions are summarized as follows:

1 The addition of Al increases the strength and hardness and improves oxidation resistance by suppressing the formation of a Mo-rich layer, reducing stress, and alleviating cracking, but Al addition will result in excessive brittleness and reduce the initial oxidation temperature at the same time.
2 Alloys with a moderate Al content of 11.1 at.% exhibit good oxidation resistance properties without a substantial decrease in mechanical performance.
3 Mixed oxide structures instead of layered oxide structures are found in the oxidized alloys, and the added Al does not form a protective Al_2O_3 layer, resulting in poor oxidation resistance.
4 Mo has the fastest diffusion rate during the oxidation process, and its velocity at the metal/oxide interface induces a porous structure and internal stress between the metal and oxide and causes spallation of the outer oxide layer.
5 The content of Mo needs to be adjusted to improve the oxidation resistance of RHEAs.

Author statement

Xiaomeng Yang, Zibing An, **Yadi Zhai** and Xin Wang: Alloy preparation and procession, Data curation, writing- original draft preparation. **Yanhui Chen and Shengcheng Mao**: Superviaion and writing-reviewing and editing, **Xiaodong Han**: Supervision.

Declaration of Competing Interest

The authors report no declarations of interest.

Acknowledgements

This work was supported by the Basic Science Center Program for Multiphase Media Evolution in Hypergravity of the National Natural Science Foundation of China (No. 51988101), Natural Science Foundation of China (Nos. 91860202, 52071003), the Beijing Natural Science Foundation (No. 1182005 and No. Z180014) and the "111" Project under the DB18015 grant and Beijing Outstanding Young Scientists Projects (BJJWZYJH01201910005018). The authors thank Dr. Dongchang Wu from Thermofisher Scientific Shanghai Nanoport for useful discussion and assistance with the Titan-Themis.

Appendix A. Supplementary data

Supplementary material related to this article can be found, in the online version, at doi:https://doi.org/10.1016/j.corsci.2021.109711.

References

[1] C.C. Juan, M.H. Tsai, C.W. Tsai, C.M. Lin, W.R. Wang, C.C. Yang, S.K. Chen, S. J. Lin, J.W. Yeh, Enhanced mechanical properties of HfMoTaTiZr and HfMoNbTaTiZr refractory high-entropy alloys, Intermetallics 62 (2015) 76–83.
[2] G.B. Wilks, O.N. Senkov, J.M. Scott, Mechanical properties of Nb25Mo25Ta25W25 and V20Nb20Mo20Ta20W20 refractory high entropy alloys, Intermetallics 19 (2011) 698–706.
[3] G. Dirras, J.P. Couzinié, Body-centered cubic high-entropy alloys: from processing to underlying deformation mechanisms, Mater. Charact. 147 (2019) 533–544.
[4] O.N. Senkov, J.P. Couzinié, D.B. Miracle, G. Dirras, Comprehensive data compilation on the mechanical properties of refractory high-entropy alloys, Data Brief 21 (2018) 1622–1641.
[5] O. Senkov, C.F. Woodward, Microstructure and properties of a refractory NbCrMo0. 5Ta0.5TiZr alloy, Mater. Sci. Eng. 529 (2011) 311–320.
[6] N. Guo, L. Wang, L. Luo, X. Li, Y. Su, J. Guo, H. Fu, Design, microstructure and mechanical properties of refractory MoNbHfZrTi high-entropy alloy, Mater. Des. 81 (2015) 87–94.
[7] K. Wang, Y. Chen, H. Wang, Z. Bai, C. Liu, L. Wu, Oxidation behavior of high entropy Mo25Nb25Ta25W25 alloy, Spec. Cast. Nonferrous Alloys 38 (2018) 661–665.
[8] C.H. Chang, M.S. Titus, J.W. Yeh, Oxidation behavior between 700℃ and 1300℃ CofRefractory TiZrNbHfTa high-entropy alloys containing aluminum, Adv. Eng. Mater. 20 (2018), 1700948.
[9] G.H. Meier, N. Briks, F.S. Pettit, Introduction to the High-temperature Oxidation of Metals, Cambridge University, 2006.
[10] W. Kai, C.C. Li, F.P. Cheng, K.P. Chua, R.T. Huang, L.W. Tsay, J.J. Kai, Air-oxidation of FeCoNiCr-based quinary high-entropy alloys at 700–900℃, Corros. Sci. 121 (2017) 116–125.
[11] B. Gorr, F. Muller, M. Azim, H.J. Christ, T. Müller, H. Chen, A. Kauffmann, M. Heilmaier, High-temperature oxidation behavior of refractory high-entropy alloys: effect of alloy composition, Oxid. Met 88 (2017) 339–349.
[12] B. Gorr, F. Muller, H.J. Christ, H. Chen, A. Kauffmann, M. Heilmaier, Effect of microalloying with silicon on high temperature oxidation resistance of novel refractory high-entropy alloy Ta-Mo-Cr-Ti-Al, Mater. High Temp. (2017) 168–176.
[13] T.M. Butler, K.J. Chaput, Native oxidation resistance of Al20Nb30Ta10Ti30Zr10 refractory complex concentrated alloy (RCCA), J. Alloys Compd. 787 (2019) 606–617.

[14] L.F. Su, L.N. Jia, J.F. Weng, Z. Hong, C.G. Zhou, H. Zhang, Improvement in the oxidation resistance of Nb–Ti–Si–Cr–Al–Hf alloys containing alloyed Ge and B, Corros. Sci. 88 (2014) 460–465.
[15] T.M. Butler, K.J. Chaput, J.R. Dietrich, O.N. Senkov, High temperature oxidation behaviors of equimolar NbTiZrV and NbTiZrCr refractory complex concentrated alloys (RCCAs), J. Alloys Compd. 729 (2017) 1004–1019.
[16] B. Voglewede, V.R. Rangel, S.K. Varma, The effects of uncommon silicides on the oxidation behavior of alloys from the Nb–Cr–Si system, Corros. Sci. 61 (2012) 123–133.
[17] K.C. Lo, Y.J. Chang, H. Murakami, J.W. Yeh, A.C. Yeh, An oxidation resistant refractory high entropy alloy protected by CrTaO4-based oxide, Sci. Rep. 9 (2019) 7266.
[18] J. Jayaraj, P. Thirathipviwat, J. Han, A. Gebert, Microstructure, mechanical and thermal oxidation behavior of AlNbTiZr high entropy alloy, Intermetallics 100 (2018) 9–19.
[19] T.M. Butler, M.L. Weaver, Oxidation behavior of arc melted AlCoCrFeNi multi-component high-entropy alloys, J. Alloys Compd. 674 (2016) 229–244.
[20] S. Wang, Z. Chen, P. Zhang, K. Zhang, C.L. Chen, B.L. Shen, Influence of Al content on high temperature oxidation behavior of AlxCoCrFeNiTi0.5 high entropy alloys, Vacuum 163 (2019) 263–268.
[21] O.A. Waseem, U. Auyeskhan, A combinatorial approach for the synthesis and analysis of AlxCryMozNbTiZr high-entropy alloys: oxidation behaviour, J. Mater. Res. 33 (2018) 3226–3234.
[22] W. Kai, C.C. Li, F.P. Cheng, K.P. Chu, R.T. Huang, L.W. Tsay, J.J. Kai, The oxidation behavior of an equimolar FeCoNiCrMn high-entropy alloy at 950°C in various oxygen-containing atmospheres, Corros. Sci. 108 (2016) 209–214.
[23] Y.J. Li, A. Kostka, A. Savan, A. Ludwig, Atomic-scale investigation of fast oxidation kinetics of nanocrystalline CrMnFeCoNi thin films, J. Alloys Compd. 766 (2018) 1080–1085.
[24] Y. Chen, F. Xue, C. Wang, X. Li, Q. Deng, X. Yang, H. Long, W. Li, L. Yang, A. Li, Effect of Cr on the microstructure and oxidation properties of Co-Al-W superalloys studied by in situ environmental TEM, Corros. Sci. 161 (2019), 108179.
[25] Q.Q. Ding, S.S. Xiang, J.X. Li, Z. Zhang, In-situ environmental TEM study of γ′-γ phase transformation induced by oxidation in a nickel-based single crystal superalloy, J. Alloys Compd. 651 (2015) 255–258.
[26] N.D. Stepanov, N.Y. Yurchenko, D.G. Shaysultanov, G.A. Salishchev, M. A. Tikhonovsky, Effect of Al on structure and mechanical properties of AlxNbTiVZr (x = 0, 0.5, 1, 1.5) high entropy alloys, Mater. Sci. Tech. 31 (2015) 1184–1193.
[27] B. Gorr, M. Azim, H.J. Christ, T. Mueller, D. Schliephake, M. Heilmaier, Phase equilibria, microstructure, and high temperature oxidation resistance of novel refractory high-entropy alloys, J. Alloys Compd. 624 (2015) 270–278.
[28] L.A. Giannuzzi, F.A. Stevie, Introduction to Focused Ion Beams, Springer, 2005.
[29] L.A. Giannuzzi, Introduction to Focused Ion Beams: Instrumentation, Theory, Techniques and Practice, Springer Science & Business Media, 2004.
[30] Q. Jeangros, T.W. Hansen, J.B. Wagner, R.E. Dunin-Borkowski, C. Hebert, J. Van herle, A. HesslerWyser, Oxidation mechanism of nickel particles studied in an environmental transmission electron microscope, Acta Mater. 67 (2014) 362–372.
[31] K.A. Unocic, D. Shin, R.R. Unocic, L.F. Allard, NiAl oxidation reaction processes studied in situ using MEMS-based closed-cell gas reaction transmission Electron microscopy, Oxid. Metal. 88 (2017) 495–508.
[32] P. Zhang, Y.T. Li, Z. Chen, J.Y. Zhang, B.L. Shen, Oxidation response of a vacuum arc melted NbZrTiCrAl refractory high entropy alloy at 800–1200°C, Vacuum 162 (2019) 20–27.
[33] O. Kubaschewski, B.E. Hopkins, Oxidation of Metals and Alloys, sl, Academic Press, 1953.
[34] M.S. Li, Y.H. Qian, X. Li, Volume ratio of an oxide to the metal, Corros. Sci. Prot. Technol. 11 (1999) 284–289.